

invoice. This is a repetitive task and needs decision-making based on the type of mismatch found. Hence, this task is typically done by humans.

Given the repetitiveness of the task and the need for decision-making, this is an ideal task for RPA. The RPA tool, when deployed, can isolate the mismatches and take the required action. Instead of having humans perform this repetitive task, using a robotic process automation tool will improve productivity by over 90 per cent, while achieving the highest accuracy. This is just one simple example of an RPA tool. Many such instances exist in the organisational workflow. Today, many of the business processes that were outsourced are getting transformed as a result of RPA. Business processes can either be front-end operations dealing with customers in call centres, or they can be back-end processes such as accounts payable, checking the derivatives trades, insurance claims processing, etc.

Automation of front-end operations typically involves chat bots or conversational agents. These automation processes usually require natural language processing or natural language understanding. Back-end operations can be either based on structured data or unstructured data. Structured data operations typically include data operations on various databases. Unstructured data operations typically include processing of documents, images and multi-modal data. Now, let us take a brief diversion to understand how natural language processing (NLP) can aid in RPA.

Natural language processing and RPA

According to industry experts, RPA is likely to progress through three stages of innovation. The first stage consists of dealing with static rules and structured data, with tasks such as data entry and validation getting automated. Stage 2 of RPA consists of some form of natural language processing capabilities and performing tasks such as content analytics and process automation. Stage 3 consists of advanced natural language processing and cognitive capabilities—this deals with RPA systems capable of human-like decision-making. Figure 1 shows the evolution of RPA capabilities through these three stages*.

While existing RPA solutions work on the 'click and automate' mode when dealing with structured data, with regard to unstructured text data, RPA has not been exploited fully for intelligent automation. Existing RPA solutions are 'action based' – they understand the actions being performed on the GUI or on the back-end enterprise systems, be it data entered in a specific field in a form or a table retrieved from a back-end database. Existing RPA scripts do not interpret the content of the textual data

that arises in the workflow. Knowledge from contextual text artefacts is not used either in process discovery or in exception management.

Moving RPA systems from the 'Repetitive Do' cycle to a 'Think – Judgement-based on context' cycle requires incorporating content analytics of textual artefacts that are present in the business process workflow. For instance, the 'Repetitive Do' cycle automation that handles structured data and static rules is better suited for automating tasks such as claim processing, account data reconciliation or data consolidation and validation. However, in scenarios where contextual knowledge and judgement are required, such as in service desk incident resolution, complaint resolution and management, eligibility processing—all of which involve unstructured text and considerable exceptions—existing RPA solutions do not fare well, due to lack of intelligent text processing.

Let us illustrate this with a simple use-case—the business workflow for a customer service desk ticket resolution system for an ecommerce retailer back-end. The system's tickets can be raised through customer chats/ emails and can either be responded to in real-time by agents via chat, or offline over emails. For each problem reported, the information is hidden in unstructured text. Each complaint needs to be identified for its respective category – such as missing item, delayed delivery, cancelling of purchase, change of address of the customer, etc. For each specific problem, there is a sequence of actions involved. All problem tickets have the initial sequence steps of:

- Authenticating the customer.
- Processing order details and retrieving them.
- Verifying the customer address, which is part of the back-end compliance mechanism.

After these steps, the problem resolution consists

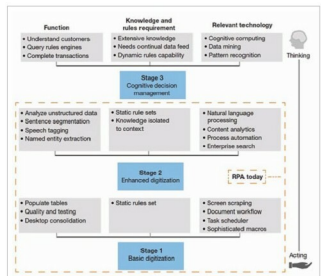


Figure 1: RPA capabilities in three stages

*As per a research report from Forrester Research on the state of RPA